AMTI (NMTC) - 2017

GAUSS CONTEST - PRIMARY LEVEL

PART - A

1. The price of an item is decreased by 25%. The percentage increase to be done in the new price to get the original price is

(A) 25 % (B) 30 % (C) 43 % (D) $33\frac{1}{2}$ %

2. How many pairs of positive integers are there such that their sum is 528 and their HCF is 33 ?

(A) 4 (B) 6 (C) 8 (D) 12

3. If one–fifth of two–third of three fourth of a number is 43, then the number is

(A) 256 (B) 540 (C) 380 (D) 430

4. The average of 8 numbers is 99. The difference between the two greatest numbers is 18. The average of the remaining 6 numbers is 87. The greater number is

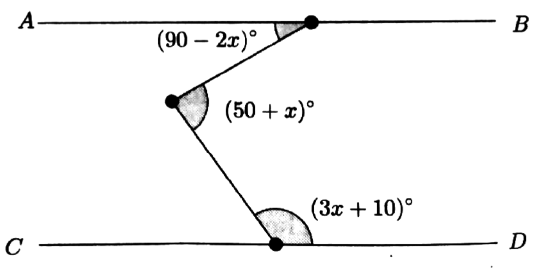
(A) 138 (B) 140 (C) 144 (D) 155

5. In a rectangle, the length is increased by 40% and breadth decreased by 30%. Then the area is

(A) increased by 5% (B) decreased by 2%

(C) decreased by 5% (D) increased by 2%

6. In the adjoining figure, lines AB and CD are parallel. What is the value of x in degrees ?



(A) 25^0 (B) 35^0 (C) 45^0 (D) 55^0

7. Find the least among the fractions $\frac{5}{6}, \frac{6}{7}, \frac{7}{8}$.

(A) $\frac{5}{6}$ (B) $\frac{6}{7}$ (C) $\frac{7}{8}$ (D) $\frac{6}{7}$ and $\frac{7}{8}$

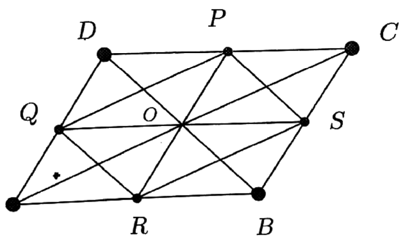
8. By how much is 15% of 23.5 more than 20% of 16.

(A) 0.125 (B) 0.325 (C) 1.5 (D) 0.235

9. How many times does the digit 1 appear when you write numbers 1 to 399 consecutively ?

(A) 180 (B) 175 (C) 178 (D) 179

10. The total numbers of parallelogram of different dimensions in the adjoining figures is



(A) 14 (B) 16 (C) 18 (D) More

PART - B

11. Consider the sequence $\frac{3}{5}, \frac{6}{7}, 1, 1\frac{1}{11}, \ldots$ The 2016th term of this sequence is $\frac{p}{q}$ where p, q are integers having no common factors, the value of q – p is ______.

12. The number of 3 digit numbers that contain 7 as at least one of the digits is ______.

13. Mahadevan conducted a problem solving session for a group of 18 primary class students. Seeing the graded performance, he distributed packets of bisecuits to all the students.

14. Using the digits of the number 2016, two digit numbers of different digits are formed. The sum of all these numbers is _______.

15. The least multiple of 7, that leaves a remainder 4 when divided by 6, 9, 15 and 18 is ____.

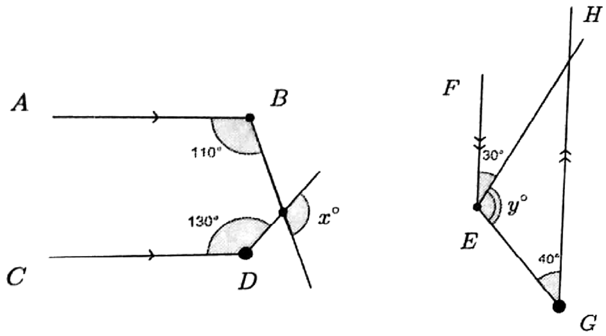
16. The number of revolutions that a wheel of diameter $\frac{7}{11}$ meter will make in going 8 kilometers on a level road is _______.

17. The radius of a circle is increased so that its circumference increases by 5%. The area of the circle will increase (in %) by _______.

18. The sum of seven numbers is 235. The average of the first three is 23 and that of the last three is 42. The fourth number is ______.

19. The number of $\frac{1}{6}$ that are in $116\frac{2}{3}$ is ______.

20. In the figure below, AB is parallel to CD and EF is parallel to GH. The value of $x^0 - y^0$ is ______.



- - - - - - - - - - - - - -

AMTI (NMTC) - 2017

GAUSS CONTEST - PRIMARY LEVEL

(Standard - V & VI)

Note :

1. Fill in the response sheet with your Name, Class and the institution through which you appear in the specified places.
2. Diagrams are only visual aids; they are NOT drawn to scale.
3. You are free to do rough work on separate sheets.
4. Duration of the test : 2 pm to 4 pm – 2 hours.

PART - A

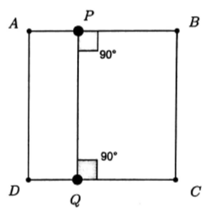
Note :

- Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.
- For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{2}$ mark.

1. Which one of the following numbers is NOT the sum of two prime numbers ?

(A) 24 (B) 30 (C) 67 (D) 21

2. ABCD is a square and PB = 2AP. The perimeter of the rectangle APQD is 80 cm. The perimeter of ABCD in cms is



(A) 100 (B) 120 (C) 140 (D) 160

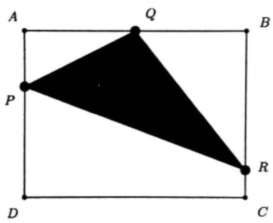
3. Saket added up all the even numbers from 1 to 101. Then, from the total he obtained, he substracted all odd numbers between 0 and 100. The answer he would have obtained is

(A) 0 (B) 20 (C) 30 (D) 50

4. The value of $\dfrac{\frac{1}{2} + \frac{1}{4} + \frac{1}{8}}{2 + 4 + 8}$ is

(A) 16 (B) 4 (C) $\frac{1}{4}$ (D) $\frac{1}{16}$

5. ABCD is a rectangle. AB = 8 cm and BC = 6 cm. Q is the midpoint of AB. P, R are on AD and BC respectively such that AP = 2 cm, CR = 1 cm. Area of the shaded triangle is square cms is

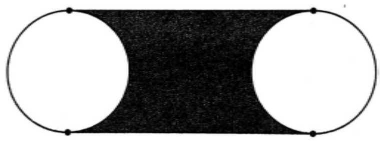


(A) 12 (B) 13 (C) 14 (D) 16

6. The Rishimoolam of a number is defined as follows. Consider the number 234. By multiplying its digits 2, 3 and 4, we obtain 2 x 3 x 4 = 24. Again, multiplying the digits of 24, we get 2 x 4 = 8. We say 8 is the Rishimoolam of the number 234. If 0 is the Rishimoolam, we say the number has no Rishimoolam. Which one of the following has no Rishimoolam ?

(A) 736 (B) 647 (C) 831 (D) 619

7. Two circles touch two parallel lines as shown in the diagram. The radius of each circle is 1 cm. The distance the centres of the circles is 5 cm. The area of the shaded region in square cms is



(A) 5π (B) 10π (C) $10 - \pi$ (D) $10 + \pi$

8. Samrud wrote two consecutive integers, one of which ends in a 5. He multiplied both. He squared the answer. The last two digits of his answer is

(A) 50 (B) 40 (C) 10 (D) 00

9. Vishwa wrote a number on each side of 3 cards. In each card, the numbers written on the sides are different. One side of each card is a prime number and the other sides had 44, 59 and 38 respectively. Given that the sum of the numbers on each card is the same, the difference between the largest and the second largest of the prime numbers on the cards is

(A) 6 (B) 7 (C) 9 (D) 4

10. The number of three digit numbers abc such that a $\times$ b $\times$ c = 15 is

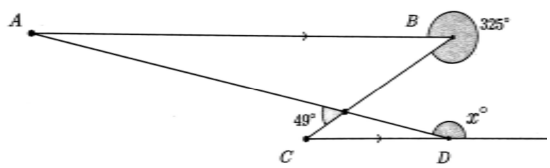
(A) 2 (B) 6 (C) 8 (D) 9

<u>PART - B</u>

Note :

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{4}$ mark.

11. Five chairs cost as much as 12 desks, 7 desks cost as much as 2 tables and 3 tables cost as much as 2 sofas. If the cost of 5 sofas is Rs. 5250, then the cost of a chair (in Rs) is ________.
12. The average age of a class of 20 children is 12.6 years. 5 new children joined with an average age of 12.2 years. The new average of the class (to one decimal place) ______.
13. 13 is a two digit prime and when we reverse its digits, the number 31 obtained is also a prime number. The number of two digit numbers living this property is ________.
14. In a garden there are two plants. One plant is 44 cm tall and the other is 80 cm tall. The first plant grows 3 cm in every 2 months and the second 5 cm is every 6 months. The number of months after which the two plants will have equal height is ________.
15. In 5 days a man walked a total of 85 KM. Every day he walked 4 KM less than the previous day. The number of KM he walked on the last day is ________.
16. In the adjoining figure, AB is parallel to CD. The value of *x* is ________.



17. In Mahadevans cycle shop for children, there are unicycles, having only one wheel, bicycles, having two wheels and tricycles, having three wheeels. Samrud counts the seats and wheels and finds that there are totally 7 seats and 13 wheels. The number of bicycles is more than tricycles. The number of unicycles in the shop is ________.
18. There is a tree with several branches. Many parrots came to rest on the tree. When 6 parrots sat on each branch of the tree, all the branches were occupied but three parrots were left over. When 9 parrots sat on each branch, all parrots were seated but two branches were empty. If b is the number of branches and p is the number of parrots, the value of b + p is ________.
19. The income of A and B are in the ratio 3:2. Their expenditures are in the ratio 5:3. If each saves Rs. 10,000, then As income is (in Rs.) ________.
20. The radius of a circle is increased so that its circumference is increased by 5%. The area of the circle will increase by _______ %.